# Calculus I

**Tutorial #7** (11 - 15 Nov 2024)

**Topics:**

- Limits of finite sums
- Definite integrals
- The Fundamental Theorem of Calculus
- Substitution method
- Areas Between Curves

**Problem 1**. For functions

$$f(x) = x^2 - 1, \quad [0, 2] \quad \text{and} \quad f(x) = \sin(x) + 1, \quad [-\pi, \pi],$$

graph each function $f(x)$ over the given interval. Partition the interval into four subintervals of equal length. Then add to your sketch the rectangles associated with the Riemann sum $\sum_{k=1}^{4} f(c_k)\Delta x_k$, given that $c_k$ is

(a) the left-hand endpoint,

(b) the right-hand endpoint,

(c) the midpoint

of the $k^{\text{th}}$ subinterval. (Make a separate sketch for each set of rectangles.).

**Problem 2**. For the function $f(x)$, find a formula for the Riemann sum obtained by dividing the interval $[a, b]$ into $n$ equal subintervals and using the right-hand endpoint for each $c_k$. Then take a limit of these sums as $n \to \infty$ to calculate the area under the curve over $[a, b]$.

$$f(x) = x^2 + 1, \quad [0, 3].$$

**Problem 3**. Graph each of functions and find its average value over the given interval.

a) $f(x) = (x - 1)^2$ on $[0, 3]$

b) $g(x) = |x| - 1$ on

$(i)$ $[-1, 1]$, $(ii)$ $[1, 3]$, $(iii)$ $[-1, 3]$.

**Problem 4**. Evaluate the integrals

a) $\displaystyle\int_{-1}^{1} (x^2 - 2x + 3)\, dx.$

b) $\displaystyle\int_{0}^{\pi} (1 + \cos(x))\, dx.$

**Problem 5**. Guess an antiderivative for the integrand function. Validate your guess by differentiation and then evaluate the given definite integral.

(*Hint:* Keep in mind the Chain Rule in guessing an antiderivative.)

a) $\displaystyle\int_{0}^{\sqrt{\pi/2}} x\cos(x^2)\, dx.$

b) $\displaystyle\int_{2}^{5} \frac{x\, dx}{\sqrt{1+x^2}}.$

**Problem 6**. Find the derivative.

(a) by evaluating the integral and differentiating the result.

(b) by differentiating the integral directly.

$$\frac{d}{dt}\int_{1}^{\sqrt{t}} \left(x^4 + \frac{3}{x^3}\right)\, dx.$$

**Problem 7**. Evaluate the indefinite integrals by using the given substitutions to reduce the integrals to standard form.

(a) $\displaystyle\int \frac{(1+\sqrt{x})^{1/3}}{\sqrt{x}}\, dx, \quad u = 1 + \sqrt{x}.$

(b) $\displaystyle\int \frac{9r^2\, dr}{\sqrt{1-r^3}}, \quad u = 1 - r^3.$

**Problem 8**. Find the areas of the regions enclosed by the curves and lines .

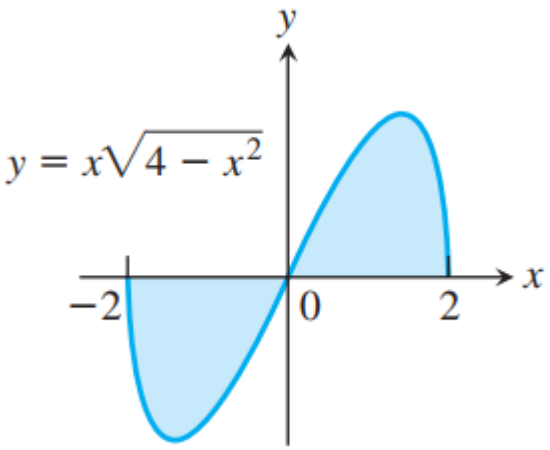
a) $x = 2y^2$, $x = 0$, $y = 3$.

b) $y^2 = 4x$, $y = 4x - 2$.

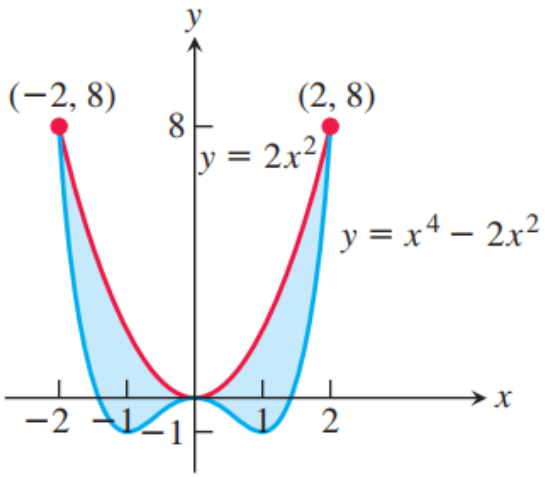
c) $y = \sin(x)$, $y = x$, $0 \le x \le \pi/4$.

Find the total areas of the shaded regions in Problems 9–10.

**Problem 9**.




**Problem 10**.




THE END

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**Answer to selected problems**

2. 12

3. (b) (i) $-\frac{1}{2}$

5. $\frac{1}{2}$, (b) $\sqrt{26} - \sqrt{5}$

7. (a) $\frac{3}{2}(1 + \sqrt{x})^{4/3} + C$, (b) $-6\sqrt{1 - r^3} + C$

9. $\frac{16}{3}$

10. $\frac{128}{15}$